eVTOL vs. eSTOL Route Comparison Tool Documentation

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Abstract

This project develops a MATLAB App Designer tool for comparing electrical vertical takeoff and landing (eVTOL) aircraft and electrical short takeoff and landing (eSTOL) aircraft across routes defined by the user. The tool uses three degree of freedom (3DOF) longitudinal equations of motion and rotor momentum theory to model the flight of the aircraft. MATLAB's ode45 was used to solve differential equations and trapezoidal integration was used to model the energy profile of the aircraft throughout the route. The comparison between the eVTOL and eSTOL aircraft relies on a normalized scoring framework across multiple components of the mission, which will be highlighted in this document.

1. INTRODUCTION

Urban and regional air mobility (UAM) has emerged as a rapidly developing sector of aerospace transportation, aiming to move passengers and cargo over short distances using electrically powered aircraft operating outside traditional airport infrastructure. Technological advancements have enabled new aircraft concepts optimized for short range operations in and around cities.

Two emerging vehicle designs are at the forefront of UAM: **electric vertical takeoff and landing (eVTOL)** aircraft and **electric short takeoff and landing (eSTOL) aircraft**. While both rely on electric propulsion systems and target similar mission profiles, they represent fundamentally different design philosophies.

eVTOL aircraft are designed to take off and land vertically, eliminating the need for runways and allowing operations from small landing pads. This infrastructure independence provides strong operational flexibility in dense urban environments where runway access is limited. However, vertical flight requires large amounts of power to generate lift purely through rotor thrust, resulting in significant energy consumption during hover and transition phases.

In contrast, eSTOL aircraft rely on aerodynamic lift from wings and require only a short runway for takeoff and landing. These aircraft use wings with high aspect ratios and distributed electric propulsion to achieve high lift at low speeds. Because the wings support the vehicle weight during most of the flight, eSTOL aircraft generally achieve higher aerodynamic efficiency and lower energy consumption during cruise. The tradeoff is that they require prepared landing strips and therefore cannot operate from any location within a city.

The key engineering question for this tool is not which vehicle type is universally superior, but instead **which configuration performs better for a given route and set of operational constraints?**. Answering this question requires comparing the vehicles under consistent physical modeling assumptions, including flight dynamics, propulsion power requirements, mission energy consumption, and operational constraints. A simulation that allows user-defined aircraft parameters and route characteristics provides a practical way to evaluate these tradeoffs and identify the conditions under which each aircraft architecture becomes advantageous.

2. AIRCRAFT MODELS AND AERODYNAMICS

2.1. Aircraft Parameters

Both aircraft are represented as parameter structs with identical field names. The user provides primary inputs through the App Designer interface, which are validated against physical limits and used for computation.. This ensures every simulation run reflects the user's actual input and therefore no hard-coded values appear inside any physics function unless required.

The user inputs are:

- Route Distance [m]
- Cruise Altitude [m]
- Cruise Speed [m/s]
- eSTOL Runway Length [m]
- Payload [kg]
- Climb Rate [m/s]
- Hover Time [s]
- Reserve Percentage
- Battery [kWh]
- Maximum Takeoff Weight (MTOW) [kg]
- Aspect Ratio
- Wing Area [m^2]

2.2. Aerodynamic Model

Aerodynamic coefficients are computed from a parabolic drag polar parameterized entirely by user inputs. The lift curve slope uses the Helmbold equation for finite wings, which accounts for aspect ratio effects on the theoretical thin-airfoil slope of 2π per radian. The simulation also uses a parabolic drag polar and a linear lift curve, as demonstrated below:

$$C_{l\alpha} = \frac{2\pi}{1+\frac{2\pi}{\pi AR}} \text{ [Helmbold equation for incompressible flow]}$$

$$C_l = C_{l\alpha}(\alpha - \alpha_0) \text{ [linear lift curve]}$$

$$C_d = C_{d0} + kC_l^2 \text{ [parabolic drag polar]}$$

where $C_{L\alpha}$ is the lift-curve slope, AR is the aspect ratio, α is the angle of attack, α_0 is the initial angle of attack, C_d is the drag coefficient, C_{d0} is the initial drag coefficient, and k is a constant.

3. 3-DOF EQUATIONS OF MOTION

3.1. State Vector and Governing Equations

The aircraft is assumed to be a point mass in the longitudinal plane, with the state vector containing four quantities:

$$x = [V, \gamma, h, x]$$

where V is the airspeed [m/s], γ is the flight path angle [rad], h is the altitude [m], and x is the ground range [m]. The governing equations of motion are derived from Newton's second law applied along and perpendicular to the flight path:

$$\frac{dV}{dt} = \frac{Tcos(\alpha)-D}{m} - gsin(\gamma) \text{ [Velocity Equation]}$$

$$\frac{d\gamma}{dt} = \frac{Tsin(\alpha)+L-mgcos(\gamma)}{mV} \text{ [Flight Path Angle Equation]}$$

$$\frac{dh}{dt} = Vsin(\gamma) \text{ [Altitude Change]}$$

$$\frac{dx}{dt} = Vcos(\gamma) \text{ [Horizontal Position Change]}$$

These equations are integrated using MATLAB's ode45 solver with relative tolerance $1 \cdot 10^{-6}$ and absolute tolerance $1 \cdot 10^{-8}$.

3.2. Quasi-static Assumption

Instead of integrating pitch dynamics as a separate state, we can compute α directly from the required lift:

$$C_{lreq} = \frac{mgcos(\gamma)}{0.5\rho V^2 S}, \alpha = \frac{C_{lreq}}{C_{l\alpha}} + \alpha_0$$

This assumption is valid for the gradual maneuvers of UAM missions since typical pitch rates during climb and cruise are well below the limit where dynamic pitch effects are significant. It also reduces the state vector from six to four variables, which considerably simplifies the design and integrator.

3.3. Thrust

Thrust is computed dynamically rather than as a fixed command, using a proportional flight path angle controller:

$$T = T_{cruise} + K_p(\gamma_{target} - \gamma)mg$$

T_{cruise} is the drag at cruise conditions because of equilibrium at level flight. The controller correction term drives gamma toward the target flight path angle throughout integration.

4. PROPULSION AND ENERGY MODEL

4.1. Hover

In hover, the eVTOL rotor system must support the full aircraft weight with zero airspeed. Ideal induced power is derived from actuator disk (momentum) theory:

$$T = (MTOW)g \text{ [Thrust in hover = Weight]}$$

Using thrust, induced velocity can be calculated using the equation: $v_i = \sqrt{\frac{T}{2\rho A}}$

Using thrust and induced velocity, the induced ideal power is: $P_{ideal} = Tv_i$, and shaft power can be calculated using an assumed field of merit (FOM) of 0.7, which is typical in hover conditions. eSTOL aircraft have no hover phase, which is very significant because the eVTOL hover condition at takeoff and landing is the primary driver of its higher energy consumption on short routes.

4.2. Transition Flight

The transition phase is unique to eVTOL and represents the window during which the aircraft accelerates from hover to cruise. During this period, lift gradually shifts from the rotor system to the wings as forward velocity increases. Wing lift grows with dynamic pressure according to:

$$L_{wing} = \frac{1}{2}\rho V^2 S C_L$$

The rotor therefore supplies only the remaining portion of the aircraft weight:

$$W_{rotor} = \max(0, W - L_{wing})$$

To estimate rotor power during forward flight, the model uses a Glauert inflow approximation. The induced velocity through the rotor disk is:

$$v_i = \sqrt{\frac{W_{rotor}}{2\rho A}}$$

and the effective velocity at the rotor disk combines induced and forward flow components:

$$V_{eff} = \sqrt{v_i^2 + \left(\frac{V}{2}\right)^2}$$

The corresponding shaft power required during transition is:

$$P_{trans} = \frac{W_{rotor}, V_{eff}}{FM \cdot \eta_{motor}}$$

As airspeed increases, the wings carry progressively more lift and rotor power decreases until the aircraft reaches the point where the vehicle transitions to fully wing-borne flight.

4.3. Propeller Actuator Disk Model

During cruise and climb the aircraft is fully wing-borne and propellers primarily provide thrust to overcome aerodynamic drag. In steady level flight the thrust requirement equals drag:

$$T_{cruise} = D$$

For climbing flight, additional thrust is required to overcome the component of weight along the flight path:

$$T = D + mg \sin\gamma$$

Propeller shaft power is computed using the actuator disk power relationship:

$$P_{shaft} = \frac{TV}{\eta_{motor}\eta_{prop}}$$

where V is the aircraft velocity, η_{motor} is the motor efficiency, and η_{prop} is the propeller efficiency. This formulation applies to both aircraft types during all non-hover phases of the mission.

4.4. Energy Integration and Battery SoC

Total mission energy is obtained by integrating propulsion power over time for each mission phase. Numerical integration is performed using the trapezoidal rule:

$$E_{phase} = \int P(t), dt \approx \text{trapz}(t, P)$$

Total mission energy is the sum across all phases:

$$E_{total} = \sum E_{phase}$$

Battery state of charge (SoC) at the end of the mission is then computed as:

$$SoC = 1 - \frac{E_{total}}{E_{battery}}$$

A feasibility constraint is applied such that the final state of charge must remain above the reserve specified by the user. Missions that violate this constraint are flagged as infeasible and excluded from scoring.

5. TRANSITION OPTIMIZATION

The eVTOL transition phase is formulated as a trajectory optimization problem in which the objective is to minimize transition energy consumption. The decision variables are incremental changes in velocity:

$$\Delta V = [\Delta V_1, \Delta V_2, ..., \Delta V_{N-1}]$$

from which the speed profile is constructed:

$$V_i = V_{start} + \sum_{k=1}^{i} \Delta V_k$$

The optimization minimizes the integrated transition power:

$$E = \int P_{trans}(V(t)), dt$$

subject to constraints enforcing a monotonic increase in speed and requiring the final velocity to reach the prescribed cruise transition speed. The problem is solved using MATLAB's *fmincon* with the Sequential Quadratic Programming (SQP) algorithm.

The optimized solution results in a profile that hovers and then accelerates, minimizing time spent in inefficient mid-transition conditions, reducing transition energy consumption relative to a simple linear acceleration profile.

However, eSTOL aircraft score zero transition penalty by definition, because they don't require a transition phase. This leads to a better score in the energy category because the entire eVTOL transition energy cost, including the hover-then-accelerate profile, is a structural consequence of the vertical takeoff requirement.

6. ALTITUDE HOLD AUTOPILOT

A simple proportional altitude controller is implemented to regulate flight altitude during simulation. Altitude error is converted into a commanded flight path angle:

$$\gamma_{cmd} = K_{p,h}(h_{ref} - h)$$

where $K_{p,h}$ is the altitude control gain. The commanded flight path angle is then tracked by the thrust controller during integration.

To represent the finite time required for the aircraft to pitch to a new attitude, a first-order lag term is included in the altitude dynamics:

$$\dot{h} = V \sin(\gamma_{cmd}) - \tau\dot{h}$$

where τ is the pitch response time constant. This prevents unrealistically instantaneous pitch changes and improves simulation stability.

7. SCORING

7.1. Metrics and Categories

Scoring for this simulation is done across eight different metrics, with each metric having its respective reference range for normalization and weight % contribution to the total score:

Metric	Reference Range	Weight %
Energy per km [kWh/km]	0.3-3.0	22
Peak Power [kW]	500-800	13
SoC at Landing [%]	20-100 %	12
Mission Time [s]	300-2400	8
Peak L/D	5-25	12
V_best_range [m/s]	20-80	8
Effective Distance [km]	d to 1.25d	13
Total Energy [kWh]	5-200	12

All the metrics are derived from the simulation results, and user inputs can flip any metric in favor of either the eSTOL or eVTOL aircraft. There are no predetermined metrics based on the different configurations of each aircraft, as this tool assumes the route configuration is accessible for each aircraft.

Each of these metrics is categorized into one of four categories, each of which has a different weightage on the overall score:

- **Energy**: The Energy per km [kWH/km] and the Peak Power [kW] metrics belong in the Energy category. This category has a overall weightage of **35%** on the overall score.
- **Performance**: The SoC at Landing [%] and the Mission Time [s] metrics belong in the Performance category. This category has a overall weightage of **20%** on the overall score.

- **Aerodynamics**: The Peak L/D and the V_best_range [m/s] metrics belong in the Aerodynamics category. This category has a overall weightage of **20%** on the overall score.
- **Operational**: The Effective distance [km] and the Total Energy [kWh] metrics belong in the Energy category. This category has a overall weightage of **25%** on the overall score.

7.2. Calculating Scores

Before any score calculation begins, a feasibility test is checked to see if either aircraft cannot complete the route above the battery reserve. If that results to be true, the aircraft score is set to zero.

Using the reference ranges provided in the table above, the values obtained for each metric are normalized to a value in between 0 and 1. The normalization against reference ranges rather than against each other means scores reflect absolute performance on a [0,1] scale regardless of how similar or different the two aircraft are. The formulas for normalization depend on the higher scoring direction of the metric.

When a lower score indicates greater efficiency, the normalization is done using:

$$n(v) = 1 - \frac{v - v_{min}}{v_{max} - v_{min}}$$

On the other hand, when a higher score indicates a greater efficiency, the normalization is done using:

$$n(v) = \frac{v - v_{min}}{v_{max} - v_{min}}$$

where v is the obtained value and (v_{min}, v_{max}) is the reference range for the respective metric. The total score for each aircraft is then summed up across all the metrics using their respective weightage (w_i):

$$Score = \sum w_i n_i$$

Below is a list of how each metric is calculated before normalization:

- **Energy per km [kW/km] =** $\frac{E_{total}}{d_{eff}}$ where E_{total} is the total mission energy consumption and d_{eff} is the effective route distance. A lower value leads to greater efficiency.

- **Peak Power [kW] =** $max(P(t))$ where $P(t)$ is the power of the aircraft as a function of time. For an eVTOL aircraft, this peak typically occurs around hover. A lower value leads to greater efficiency.

- **SoC at Landing [%] =** $1 - \frac{E_{total}}{E_{battery}}$ where E_{total} is the total mission energy consumption and $E_{battery}$ is the battery energy consumption. A lower value leads to greater efficiency.

- **Mission Time [s] =** t_{final}, the time it takes for the mission to complete. A lower value leads to greater efficiency.

- **Peak Lift to Drag Ratio (L/D) =** $\frac{C_l^*}{C_{d0} + kC_l^2}$, where $C_l^* = \sqrt{\frac{C_{d0}}{k}}$. This is derived from the parabolic drag polar, where at the peak L/D, $\frac{dC_d}{dC_l} = 0$. A higher value leads to greater efficiency.
- V_best_range **[m/s] =** $\sqrt{\frac{2(MTOW)g}{\rho S C_l^*}}$ where $(MTOW)$ is the maximum takeoff weight of the aircraft, g is the standard gravitational acceleration, ρ is the standard air density, S is the wing area, and $C_l^* = \sqrt{\frac{C_{d0}}{k}}$. At best range speed, lift is equal to weight and $C_l = C_l^*$. Therefore, the equation for V_best_range is derived from equating lift to weight: $\frac{1}{2}\rho V^2 S C_l^* = (MTOW)g$ and solving for V. A lower value leads to greater efficiency.

- **Effective Distance [km] =** d_{route} for an eVTOL aircraft that doesn't have any runway offset. For an eSTOL aircraft, $d_{eff} = d_{route} + d_{offset}$, where d_{route} is the total route distance and d_{offset} is the runway offset distance.

- **Total Energy [kWh] =** E_{total}, which was calculated by numerically integrating the power function. A lower value leads to greater efficiency.

8. LIMITATIONS

This model simplifies several aspects of aircraft performance and urban air mobility operations. First, the propulsion model uses actuator disk momentum theory, which captures the main physics of rotor and propeller power but does not include detailed aerodynamic effects such as blade element aerodynamics, rotor–wing interference, or unsteady transition dynamics. As a result, the calculated power requirements should be interpreted as approximate estimates rather than exact performance predictions.

Second, the aircraft are modeled using simplified aerodynamic parameters such as a constant lift coefficient and fixed lift-to-drag ratio. Real aircraft experience variations in aerodynamic performance with angle of attack, Reynolds number, and configuration changes during flight. These effects are not included in the current simulation.

The transition phase for the eVTOL aircraft is also simplified using an approximate inflow model and an optimized velocity profile. In reality, transition flight is a complex process involving control system dynamics, rotor tilting, and aerodynamic interactions between rotors and wings. These factors may affect the true energy and power requirements of the vehicle.

Additionally, the scoring framework relies on user-defined normalization ranges and category weights. While this allows flexibility in evaluating different mission priorities, the final score can vary depending on the selected weights and reference ranges. The results therefore represent a relative comparison under the chosen assumptions rather than a universal ranking of aircraft types.

Finally, the model does not include several real-world operational considerations such as weather conditions, air traffic constraints, passenger loading, noise limitations, or infrastructure availability. These factors can significantly influence the feasibility and efficiency of urban air mobility operations.

9. CONCLUSION

This project developed a simulation framework to compare the performance of eVTOL and eSTOL aircraft for urban air mobility missions. The model simulates aircraft trajectories across multiple flight phases including hover, transition, climb, cruise, and descent while estimating propulsion power, total energy consumption, and mission performance.

Based on simulation results, eSTOL aircraft generally achieve lower energy consumption and lower peak power requirements due to their ability to operate primarily in efficient wing-borne flight. In contrast, eVTOL aircraft require significant power during hover and transition phases, which increases both peak power demand and total mission energy usage.

However, eVTOL aircraft provide operational advantages because they can operate from smaller landing areas and do not require runways. This flexibility can reduce effective travel distance and allow operations in locations where conventional runway infrastructure is unavailable.

To compare the aircraft fairly across multiple performance aspects, a scoring framework was implemented. This framework evaluates each aircraft using metrics related to energy efficiency, performance, aerodynamics, and operational factors. The scoring system provides a measurable method to evaluating aircraft suitability for different mission conditions and route characteristics.

Overall, the analysis demonstrates that the relative advantages of eVTOL and eSTOL aircraft depend strongly on mission requirements. eSTOL aircraft tend to perform better in terms of energy efficiency and aerodynamic performance, while eVTOL aircraft offer greater operational flexibility in dense urban environments. The simulation framework developed in this project provides a useful tool for exploring these tradeoffs and evaluating aircraft performance across a range of urban air mobility scenarios.

10. SOFTWARE

All simulation code is available on GitHub. The project is structured as listed below:

- *constants.m* - This script contains physical constants and the required solver settings and tolerances.
- *build_aircraft.m* - This script validates the user input and computes the required derived quantities.
- *eom_3dof.m* - This script uses the 3-DOF longitudinal equations of motion and the ode45 solver and uses the quasi-static assumption for the computation of lift and drag.
- *thrust_model.m* - This script computes the thrust and shaft power during hover, transition, and cruise. It uses the derived equations for the three phases of the mission.
- *energy_model.m* - This script computes the energy over a simulated flight phase using trapezoidal integration.
- *mission_energy.m* - This script puts everything all together by simulating a full route and returning metric values that need to be normalized for scoring.
- *transition_opt.m* - This script solves for the optimal speed profile using *fmincon* and then compares it against a linear acceleration.
- *scoring_framework.m* - This script calculates the individual and total scores for the metrics, categories, and aircraft using the developed scoring framework.
- *recommendation_engine.m* - This script prints out the text for the recommendation panel in the app.
- *routecomparisontool.mlapp* - This is the MATLAB App Designer user interface used to display the tool. The app includes a section where the user can input their parameters and displays corresponding plots and scores.

Dependencies: MATLAB R2023b+, Optimization Toolbox (fmincon), App Designer (included in base MATLAB).

11. GITHUB REPOSITORY

Visit the repository using the link below to access the source code and the app:

https://github.com/ksasank10/UAM-Route-Comparison-Tool